12 Justificați cu definiția valoarea limitelor

a) $\lim_{n\to\infty} \frac{1}{\sqrt{n}} = 0$ → am ieșit la tablă

b) $\lim_{n\to\infty} \frac{n^2}{n+1} = \infty$

$\forall M \in \mathbb{R}\ \exists N \in \mathbb{N}$ $N = [2M]+1$ pt $\forall n \geq N$

$(n-1)^2 > 0 \quad \forall n \geq 2$

$\frac{n^2}{n+1} > \frac{n}{2}$

$2n^2 > n^2 + n$

$n^2 > n$

$n > 1$

$\frac{n^2}{n+1} > \frac{n}{2} > \frac{[2M]+1}{2}$

??

b) $\lim_{n\to\infty} x_n = +\infty \Leftrightarrow \forall M \in \mathbb{R}, \exists N \in \mathbb{N}$ a.i. $\forall n > N \Rightarrow x_n > M$

DEF

$\frac{n^2}{n+1} > \frac{n}{2} \quad \forall n \geq 2 \quad (*)$

$\Leftrightarrow 2n^2 > n^2 + n$

$\Leftrightarrow n^2 > n$

$\Leftrightarrow n > 1$

Fie $M \in \mathbb{R}$, fie $N = \max\{2, [2M]+1\}$

$\Rightarrow N \geq 2$

$N \geq [2M]+1$

$\Rightarrow \forall$

Din (*)

$\Rightarrow x_n > \frac{n}{2}$

Cum $n > N \geq [2M]+1 \geq 2M$ $\quad ([x] \geq x-1)$ $\Big\} \Rightarrow$

$\Rightarrow \forall n > N \Rightarrow x_n > \frac{n}{2} \geq M \Rightarrow x_n > M$

$N = \max\{2, [2M]+1\}$

$\Rightarrow \lim\limits_{n\to\infty} x_n = +\infty$

$\boxed{3}$ Studiați convergența șirului $X_n$ și calculați limita acolo unde este pos..

a) $X_n = a^n$, $a \in \mathbb{R}$

b) $X_n = \frac{2^n}{n!}$

c) $X_n = \sqrt[n]{n}$

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Rezolvare:

UN ȘIR E CONVERGENT $\Leftrightarrow \forall \varepsilon > 0, \exists N \in \mathbb{N}$ a.î. $\forall n \geq N, |X_n - L| < \varepsilon$

a) $X_n = a^n$, $a \in \mathbb{R}$

I $\;a < -1$

$n_k = 2k$, $k \in \mathbb{N}$
$n'_k = 2k+1$

$\lim\limits_{n\to\infty} x_{n_k} = \lim$ ...

(II) $a=-1$

$\Rightarrow X_n = (-1)^n$

— " —

— " —

$\left.\begin{array}{l} X_{2k} \to 1 \\ X_{2k+1} \to -1 \end{array}\right\}$ diferite $\Rightarrow$ nu e convergent pe $-1$

(III) $a \in (-1;1)$

Fie $\varepsilon > 0$, $\exists N$? a.î. $\forall n > N \Rightarrow |X_n - 0| < \varepsilon$ ?

$|X_n - 0| < \varepsilon$

$\Leftrightarrow |a^n| < \varepsilon$

$\Leftrightarrow |a|^n < \varepsilon$

$\Leftrightarrow n \ln|a| < \ln \varepsilon$

$\Leftrightarrow n > \dfrac{\ln \varepsilon}{\ln|a|}$

Luând $N = \dfrac{\ln \varepsilon}{\ln|a|} \Rightarrow \lim\limits_{n\to\infty} X_n = 0 \Rightarrow (X_n)$ - convergent

(IV) $a = 1 \Rightarrow X_n = 1$ constant $\Rightarrow X_n$ - convergent

(V) $a > 1$

$\lim\limits_{n\to\infty} X_n = \lim\limits_{n\to\infty} a^n = \ldots = +\infty \notin \mathbb{R} \Rightarrow X_n$ - divergent

Concluzie: Dacă $a \le -1 \Rightarrow X_n$ - divergent

Dacă $a \in (-1,1] \Rightarrow X_n$ - convergent

Dacă $a > 1 \Rightarrow X_n$ - divergent.